

Homework 3

1. Find the sum for each of the following series:

(a) $\sum_{n \geq 2} \ln \left(1 - \frac{1}{n^2}\right).$

(b) $\sum_{n \geq 1} \frac{n+1}{3^n}.$

(c) $\sum_{n \geq 1} \frac{n}{n^4 + n^2 + 1}.$

2. Study the convergence of the following series:

(a) $\sum_{n \geq 1} \frac{x^n}{n^p}, x > 0, p \in \mathbb{N}.$

(c) $\sum_{n \geq 1} (\sqrt[n]{n} - 1).$

(b) $\sum_{n \geq 2} \frac{1}{(\ln n)^{\ln n}}.$

(d) $\sum_{n \geq 1} \frac{1 \cdot 3 \cdot \dots \cdot (2n-1)}{2 \cdot 4 \cdot \dots \cdot 2n} \cdot \frac{1}{2n+1}.$

3. Start with an equilateral triangle of side 1. For each side, remove the middle third and add there another equilateral triangle. Repeat this process at each iteration (see the figure). How many sides are there at iteration n ? What is the limit of the perimeter and the area?

